FACULTY OF SCIENCE AND TECHNOLOGY
UNIVERSITAS ISLAM NEGERI WALISONGO SEMARANG, INDONESIA

# CERTIFICATE OF APPRECIATION

**B-4523/UN.10.08/D/SP.01.01/07/2024**

THIS IS TO CERTIFY THAT:

## Ariko Yahya Setyawan

### Malang State University

**Scientific Papers: Innovation Start-up Category in the Information Technology**

for being "The Gold Medalist" in
the International Walisongo Science Competition 2024.

Semarang, July 27, 2024.

**Prof. Dr. H. Musahadi, M.Ag**
**Dean, Faculty of Science & Technology**

**Muji Suwarno, M.Pd.**
**Chief Committee**

# DoveComm: An Innovative Chat Application for Offline Communication in Remote Areas and Emergency Situations

Ariko Yahya Setyawan[1], Rr. Rakhel Olivia Ananda[2], Cahya Adidarma[3], Fakhru Widyan Firdaus[4]

ariko.yahya.2003226@students.um.ac.id
rr.rakhel.2103226@students.um.ac.id
cahya.adidharma.2203326@students.um.ac.id
fakhru.widyan.2103226@students.um.ac.id
State University of Malang

**Abstrack**

In today's digital era, ensuring effective communication is crucial, especially in remote areas and during emergencies where internet access is often unavailable. DoveComm, an innovative startup, addresses this challenge by developing an Android chat application that operates without internet connectivity using Wi-Fi and Amplitude Modulation (AM) technology. The application is designed to maintain communication in regions with limited internet access and during natural disasters that disrupt telecommunication infrastructure. The primary goal of DoveComm is to create a reliable and effective communication tool for use without internet access, targeting remote areas and emergencies. The objective of this study is to develop a chat application that facilitates communication in the absence of internet connectivity, leveraging Wi-Fi and AM technology. The application also integrates artificial intelligence (AI) to analyze communication patterns and provide automatic alerts to users. The methodology involves the development and testing of the application in simulated environments to evaluate its effectiveness and reliability. The process includes the integration of peer-to-peer communication based on Wi-Fi, combined with AM technology to extend the communication range. User feedback and performance metrics are used to refine the application. AI-driven chatbots are incorporated to analyze message patterns and keywords, issuing alerts about potential emergencies based on the communication content. Results from preliminary testing indicate that DoveComm effectively facilitates communication without internet connectivity, providing reliable messaging and alert capabilities in areas with limited network access. The AI-powered chatbot successfully identifies and alerts users to potential emergencies based on message patterns and keywords, enhancing coordination and response in disaster scenarios. The significance of these results lies in the substantial improvement in communication technology for areas with inadequate internet infrastructure and for disaster management. DoveComm represents a significant advancement by ensuring continuous information flow and improving safety, social connectivity, and operational efficiency in critical situations. This study contributes to the development of resilient communication systems essential for applications in remote and emergency scenarios and supports DoveComm's mission as a startup to create innovative technological solutions with a positive impact on society.

**Keywords: Startup, Offline chat application, WiFi, Amplitude Modulation, Artificial Intelligence, Emergency Communication, Remote Area Connectivity.**

## Introduction

In the current technological era, the need for fast and easy information dissemination facilities is increasing, especially with the growth of Android-based devices (Natanael et al., 2018). The rapid development in communication technology has significantly changed the way humans interact. However, despite the internet being the main medium of communication, access limitations occur in areas not covered by internet networks. This presents challenges for communities in effectively connecting and accessing information.

A mid the rapid development of information technology, many regions in Indonesia still face challenges in accessing the internet. In some areas, there is no internet access at all (Ruth, 2015). This issue is a serious barrier to reducing the digital divide between urban and rural areas. Meanwhile, the importance of effective communication is increasing, especially in emergency situations such as natural disasters.

Natural disasters often cause damage to telecommunications infrastructure, impacting internet access. In such situations, communication becomes crucial for coordinating evacuations, providing emergency aid, and saving lives. Therefore, a communication solution that can function without relying on potentially damaged or unavailable internet infrastructure is needed (Wahyuni, 2016).

In an era where wireless connectivity is becoming increasingly important in daily life, communication applications have become a dominant aspect of human activity. Wireless networks enable humans to communicate using various electronic devices such as computers, mobile phones, and other devices without using cables as the transmission medium. These wireless networks use electromagnetic signals that flow through the air, and in this context, they utilize radio waves (Maulana, 2023). Amid the rapid advancement of wireless services in daily life, the demand for high-speed wireless services is increasing. This indicates the need for innovation in wireless technology to meet consumer demands for fast connectivity that can be used in areas without internet connectivity.

Several studies have been conducted, as noted by Ekata M. Lambture and Prof. Z.M. Shaikh (Journal et al., 2016), who investigated Android chat applications based on Wi-Fi technology providing similar features but using Wi-Fi (wireless). Android chat applications based on Wi-Fi technology allow users to communicate with each other through Wi-Fi by simply activating the Wi-Fi option on their phones instead of using internet data connectivity. Thus, it saves costs as users can send and receive messages within the range of Wi-Fi. There is no need for an external connection. Within the Wi-Fi range, users can transfer large amounts of data and download data. However, a limitation in this study is the limited Wi-Fi range, which is only 150 feet or 45 meters.

Situations that disrupt telecommunications infrastructure. Using the Research and Development DoveComm, as an innovative startup, aims to address this challenge by developing an Android-based chat application that can operate without internet connectivity using Wi-Fi and Amplitude Modulation (AM) technology. This startup aims to create a reliable communication solution in areas with limited internet access and in emergency methods. This research aims to develop an innovative and adaptive chat application. This application is designed to allow individuals to communicate without internet access. By utilizing Wi-Fi and Amplitude Modulation (AM) technology, this application is expected to operate well in remote areas or in emergency situations where internet infrastructure is damaged or unavailable. Additionally, the integration of artificial intelligence (AI) for data analysis on servers and notification chatbots is expected to add functionality to the application. AI can be used for analyzing message data sent through the chat application, identifying usage patterns, and providing relevant automatic notifications. For example, AI can detect emergency situations based on keywords in messages and send notifications to authorities or rescue teams.

Therefore, this research is expected to contribute significantly to solving the problem of communication accessibility in remote areas and in emergency situations in Indonesia.

**Literature review/previous works**

Wireless technology, especially Wi-Fi, has become an integral part of modern communications. Wi-Fi allows devices to connect wirelessly over a local network. Wi-Fi standards such as IEEE 802.11ac and 802.11ax, provide high data transfer speeds and wide coverage. This technology uses radio frequencies in the 2.4 GHz and 5 GHz ranges to transmit data, thus enabling reliable and fast connectivity in various situations. (Yusantono , 2020).

Wireless network topology refers to the way devices on a wireless network are arranged and communicate with each other. Several common topologies are used in wireless networks, one of which is Ad-Hoc and infrastructure topologies. Ad-Hoc topology is a simple wireless network topology that connects two digital devices directly without an intermediary in the form of a wireless access point or router. This topology is often called a Peer-to-Peer relationship or Independent Basic Service Set (IBBS), which connects more than two digital devices through an intermediary in the form of a wireless access point or router. The access point will be the center of the network, so its function is the same as a hub or switch. This topology is divided into two, namely the Basic Service Set (BSS) and the Extended Service Set (ESS).

In BSS, digital devices are connected to one wireless access point. In ESS, digital devices are connected to more than one wireless access point. Wi-Fi was first introduced in 1999 by the United States network certification agency. Wi-Fi is a technology used in wireless networks (Amin et al., 2022). Wi-Fi using the IEEE 802.11 standard is the most commonly used wireless technology for local networks. This standard provides a variety of operating frequencies and modulation methods that allow different speeds and ranges. For example, 802.11n can operate at 2.4 GHz and 5 GHz frequencies using Multiple Input Multiple Output (MIMO) techniques to increase data throughput (Wireless, 2016). The maximum Wi-Fi coverage distance is approximately 30 meters from the transmitter (Access Point). Wi-Fi can be divided into several types based on IEEE standards.

| Standar IEEE | Frekuensi | Kecepatan Maksimum | Kompatibel |
|---|---|---|---|
| 802.11 | 2,4 GHz | 2 Mbps | - |
| 802.11a | 5 GHz | 54 Mbps | - |
| 802.11b | 2,4 GHz | 11 Mbps | - |
| 802.11g | 2,4 GHz | 54 Mbps | 802.11b |
| 802.11n | 2,4 GHz dan 5 GHz | 600 Mbps | 802.11a/b/g |
| 802.11ac | 5 GHz | 1,3 Gbps | 802.11a/n |
| 802.11ad | 2,4 GHz, 5 GHz dan 60 GHz | 7 Gbps | 802.11 a/b/g/n/ac |

Table 2.2.1 Wi-Fi Based on IEEE Standards

Amplitude modulation is often used in wireless communications applications such as AM radio broadcasts, aviation communications and walkie-talkie communications. Despite its simplicity, AM has advantages in simplicity of implementation and compatibility with existing infrastructure. However, its main drawback is its low spectral efficiency because it requires a larger bandwidth. WiFi-based chat applications allow users to communicate via a local network without requiring an internet connection. This is especially useful in areas without reliable internet access. This application can improve communications in emergency situations, such as natural disasters, where conventional telecommunications infrastructure may not work.

This proposed chat messenger connects you instantly with people on the same Wi-Fi network without ever accessing the Internet. Thus, it saves costs when users can send and receive messages within Wi-Fi range. However, there is no need for an external connection. Within the Wi-Fi range, users can transfer large amounts of data as well as download data.

The transceiver plays a role in converting radio signals from digital data generated by a computer or other device into analog radio signals for transmission via an antenna, and vice versa. This makes it more efficient and practical, especially in communication systems such as radio, cell phones, and computer networks. The transmitter, which functions to convert information signals in the form of digital data into analog signals which can be transmitted via media such as air or cable, a receiver which functions to receive the transmitted analog signal and convert it back into digital data which can be processed by a receiving device. This process involves modulation, where digital data is converted into an analog carrier signal. Where the transmitter will convert digital data into a signal form that can be transmitted. The signal is then amplified to ensure it can reach the recipient without significant degradation. Meanwhile, the receiver converts the analog carrier signal back into original digital data. This is the reverse of the modulation process. So, the signal can be filtered to remove unwanted interference and noise so that the signal is more relevant when processed.

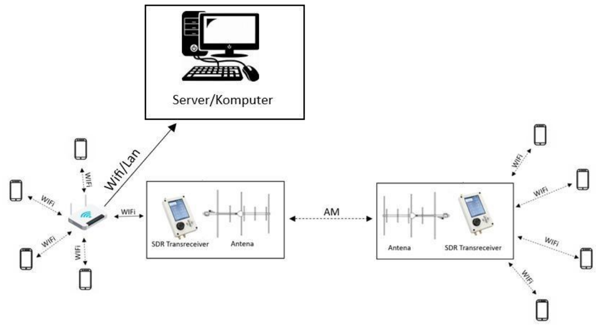
A tool that allows various devices such as computers, smartphones and tablets to connect to the internet or local network wirelessly is a Wi-Fi router. Modern Wi-Fi routers, especially those that support dual-band, offer the ability to send data over two different radio frequencies, namely the 2.4 GHz and 5 GHz bands. We need a signal whose transmission is straight and directional (directed) and emits waves at only one frequency, so we use a Yagi antenna.

Yagi antennas generally consist of a reflector, a driven element, and several directors. Yagi antennas have several advantages, including very cheap construction and high directivity. The way this antenna works is that the driven element will produce electromagnetic waves when given a signal input. Next, the reflector will reflect the waves back towards the driven element, creating constructive interference that strengthens the signal going forward. So the directors will guide the forward wave by creating constructive interference in a certain direction, resulting in a great gain in that direction.

## Methods

### Implementation and Assembly of Hardware

The first is setting up the Wi-Fi Router by configuring the Wi-Fi router for the local network which will send data to the SDR Transceiver. Secondly, set Port Forwarding, set the port forwarding for communication with the SDR Transceiver. The next step is to configure the SDR Transreceiver Sender by connecting the SDR Transreceiver to the Wi-Fi router. If the SDR Transreceiver supports Wi-Fi, set it to connect to a Wi-Fi network already configured on the router. After connecting, install the appropriate antenna on the SDR Transreceiver to ensure signal transmission effectively. To view the SDR Transreceiver Receiver Configuration by installing an antenna on the SDR Receiver to receive AM signals. Then, they set reception parameters such as frequency, AM modulation, and sensitivity. The latter by Configuring the SDR Transreceiver to convert the AM signal to Wi-Fi for Android reception.




Hardware Design Scheme

**Implementation and Configuration of Software**

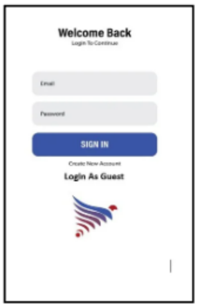
The first one is to set up the Project by creating a New Project in Android Studio: Start by creating a new project in Android Studio with the appropriate initial configuration. Next to Configure Dependencies: make sure your project has dependencies required to implement the Wi-Fi feature. This includes Android SDK dependencies and classes required for network communications.

The second is for User Interface (UI) Design by Creating XML Layouts: Design your user interface (UI) with XML in the layout file. This may include message lists, message input fields, buttons to send messages, and connection status.

Once connected Connect UI with Java Code: Using findViewById() function to connect UI elements with your Java code. Thirdly, Wi-Fi Connection Management by initializing Wi-Fi Manager: Initialize Wi-Fi manager and channels for communication. For Receiver List: Registers receivers to handle Wi-Fi events such as changes to device status or discovery.

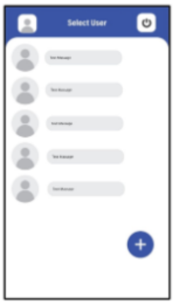
Then start Device Discovery: When the "Discover Devices" button is pressed, start discovery of Wi-Fi devices around the last one. Respond to Connection Events: Respond to Wi-Fi connection events, such as devices connected or disconnected.

Front Page Design




SIGN IN Design


SIGN UP Design




Chat Home Design

Chat Room Design

**Server Implementation for AI and Chatbot**

The first one is to set up the server, by installing AI software and chatbot modules, Then connect the server to the SDR Transreceiver. The second is for analyzing AI: observing the receipts and analyzing message data to detect emergency situations or anomalies. The third is for Chatbot Development: designing a chatbot for automatic notifications by integrating with AI analysis results.

**Results and discussion**

With its positive aspects that include offline capabilities, mesh networking, and wide deployment possibilities, DoveComm is an attractive solution to overcome communication barriers in remote areas and emergency situations. While there are some issues to consider, DoveComm remains a valuable tool for a variety of communication needs that require reliability and rapid response. DoveComm has the potential to be an effective and reliable means of communication in emergency situations that require good coordination and cooperation.

However, while DoveComm has many advantages, there are some issues to consider. Firstly, users need to ensure that their device is always charged and in good condition, as DoveComm relies on the presence of the device to communicate. In addition, users also need to understand how to properly use the device to maximize DoveComm's potential in emergency situations.

DoveComm also offers wide deployment possibilities, not only in remote areas but also in more urbanized locations. This makes it a valuable tool for a wide range of communication needs, from everyday use to emergency situations that require quick response and good coordination. With its flexible and easy-to-use features, DoveComm can be a suitable solution for any communication scenario required.

In addition, DoveComm uses a mesh networking approach that allows devices to communicate directly with each other. This helps extend network coverage beyond a single access point, which is especially useful in situations where infrastructure is damaged or limited. As such, DoveComm can be an effective tool for emergency responders, disaster relief teams, and large event organizers who require reliable and fast communication.

DoveComm is presented as an attractive solution to overcome communication barriers in remote areas and emergency situations. One of the positive aspects of DoveComm is its ability to operate offline. This allows users to stay connected even in geographically isolated areas or during power outages caused by disasters. By eliminating reliance on internet connections, DoveComm can help in coordinating rescue efforts, sharing vital information, and maintaining a sense of community.

While DoveComm's primary focus is on addressing communication barriers in remote areas, its deployment possibilities extend far beyond. The self-contained communication network created by DoveComm makes it a versatile tool that can be utilized in various settings. Whether in non-remote locations facing communication challenges or as a backup system in urban centers, DoveComm offers a reliable alternative for establishing connectivity in diverse environments. Its adaptability and effectiveness make DoveComm a valuable asset for enhancing communication capabilities across different scenarios.

The mesh networking functionality of DoveComm sets it apart by enabling direct device-to-device communication without reliance on a central access point. This approach proves invaluable in situations where traditional communication infrastructure is damaged or limited, such as during natural disasters or emergencies. By creating a decentralized network, DoveComm ensures reliable communication channels even in challenging conditions, enhancing connectivity when it matters most.

In remote areas and emergency situations, DoveComm's offline capability is a game-changer for ensuring efficient communication. By staying connected even without an internet connection, DoveComm enables seamless coordination of rescue efforts and sharing of critical information. This feature is particularly crucial in scenarios where power outages are common due to natural disasters, allowing uninterrupted communication for those in need of assistance.

**Conclusions**

The conclusion based on the results of data analysis is that the need for fast and easy communication is increasing, especially in areas where the internet is not reachable, even in emergencies such as natural disasters that often damage telecommunication infrastructure, so alternative communication solutions are urgently needed. By developing a chat application that works without internet access using Wi-Fi technology and AM modulation by Integrating artificial intelligence (AI) for automatic message and notification data analysis, it will improve connectivity in remote areas and improve the effectiveness of communication in emergency situations, so that it can detect emergency situations and provide quick notifications to users. To overcome these challenges, we conducted research entitled "Implementation of Chat Applications Using Wi-Fi with Amplitude Modulation (AM)" using the Research and Development method.

This research aims to develop innovative and adaptive chat applications. The app is designed to allow individuals to communicate without the need for internet access. By using Wi-Fi technology and Amplitude Modulation (AM) modulation, the app is expected to operate properly in remote areas or in emergency situations where internet infrastructure is damaged or unavailable. We are confident that this startup will succeed and have a significant impact because the App offers innovative solutions to communication problems in areas that do not have internet access, especially in emergency situations such as natural disasters. By using Wi-Fi and AM modulation, these applications can operate without relying on internet infrastructure that is often damaged or unavailable during disasters.

The integration of artificial intelligence (AI) in this application adds significant added value. AI is used for message data analysis, detecting emergency situations based on keywords, and providing relevant automated notifications to authorities or rescue teams, which can improve responsiveness and efficiency in disaster management. The app provides communication solutions in areas that are not covered by conventional internet networks, utilizing Wi-Fi networks and AM modulation. This will facilitate communication between individuals in remote areas, improving social cohesion and information exchange. With the ability to operate without the internet, the app enables reliable communication during emergency situations, aids in coordination between rescue teams, the government, and the community, and improves individual safety by providing reliable alternative lines of communication during disasters. The target users of this application include people in remote or isolated areas and disaster management teams. Thus, this application will not only help in emergency situations but will also provide long-term benefits in improving connectivity in areas that are underserved by conventional communication infrastructure.

**References**

Amin, M., Raja, H. D. L., Nur, M. N. A., Prasetio, A., Sulaiman, O. K., Karim, A., Muttaqin, Sihotang, J. I., Simarmata, J., Jamaluddin, Harlina, S., A, A., & Pakpahan, A. F. (2022). Teknologi Jaringan Nirkabel. In *Yayasan Kita Menulis*.

Journal, I., Engineering, O. F., Chatting, A., Based, A., & Technology, O. N. W. (2016). *International journal of engineering sciences & research technology android chatting application based on wi-fi technology *. 5*(5), 103–106.
Lathi, B. P., Ding, Z., Freq, S.-14, Ding, Z., Lathi, B. P., Chu, J., Alencar, M. S., Da Rocha, V. C., Smillie, G., Nassar, C., Stix, G., Vitetta, G. M., Taylor, D. P., Colavolpe, G., Pancaldi, F., Martin, P. A., Barué, G., Rohde, U. L., Kuhn, N., … Akaiwa, Y. (2010). and Analog Modern Digital Communication Systems. In *Telecommunications Demystified* (Vol. 41, Issue 1). http://portal.acm.org/citation.cfm?id=541365%0Ahttp://www.techniajournal.com/attachments/article/128/ANALOG & DIGITAL MODULATION.pdf
Maulana, I. (2023). Jaringan Nirkabel: Evolusi Teknologi Dan Dampaknya Pada Masyarakat. *Jurnal Teknologi Terkini*, *3*(10), 1–19.
Natanael, D., Faisal, & Suryani, D. (2018). Text Encryption in Android Chat Applications using Elliptical Curve Cryptography (ECC). *Procedia Computer Science*, *135*, 283–291. https://doi.org/10.1016/j.procs.2018.08.176
Nirkabel, S. K. (2016). *9. Sistem Komunikasi Nirkabel*.
Ruth, E. (2015). Deskripsi Kualitas Layanan Jasa Akses Internet di Indonesia dari Sudut Pandang Penyelenggara. *Buletin Pos Dan Telekomunikasi*, *11*(2), 137. https://doi.org/10.17933/bpostel.2013.110204
Wahyuni, R. A. (2016). Pemanfaatan Teknologi Informasi Dalam Penanggulangan Bencana Alam Di Indonesia Berbasiskan Web. *Bina Insani Ict Journal*, *3*(1), 1–6. http://101.255.92.196/index.php/BIICT/article/view/771%0Ahttp://101.255.92.196/index.php/BIICT/article/download/771/618
Yusantono. (2020). Analisis dan Perbandingan Jaringan WiFi dengan frekuensi 2.4 GHz dan 5 GHz dengan Metode QoS. *Journal of Information System and Technology*, *05*(05), 34–52.